

D-SCRAMBLED CANTOR SETS

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ABSTRACT. For all $d \geq 3$, we provide an example of a homeomorphism of Cantor space that has an uncountable d -scrambled set but no d -scrambled Cantor set.

INTRODUCTION

Suppose that X is a metric space and $f: X \rightarrow X$. We say that $x \in X^d$ is *proximal* if $\liminf_{n \rightarrow \infty} \max_{i < j < d} d_X(f^n(x(i)), f^n(x(j))) = 0$, *separated* if $\limsup_{n \rightarrow \infty} \min_{i < j < d} d_X(f^n(x(i)), f^n(x(j))) > 0$, and *Li–Yorke* if it is proximal and separated. A set $Y \subseteq X$ is *d -scrambled* if every injective sequence in Y^d is Li–Yorke, and f is *Li–Yorke d -chaotic* if there is an uncountable d -scrambled set $Y \subseteq X$.

In [GGM21], it was shown that every Li–Yorke 2-chaotic continuous function on a Polish space has a 2-scrambled Cantor set. Here we show:

Theorem 1 (ZFC). *Suppose that $d \geq 3$. Then there is a Li–Yorke d -chaotic homeomorphism of $2^{\mathbb{N}}$ with no d -scrambled Cantor set.*

A *graph* on X is an irreflexive symmetric set $G \subseteq X \times X$. Let $\text{Hyp}^d(G)$ be the d -ary relation on X of all injective sequences $x \in X^d$ such that $G \cap (x[d] \times x[d]) \neq \emptyset$. An *embedding* of a d -ary relation R on X into a d -ary relation S on Y is an injection $\pi: X \rightarrow Y$ such that $x \in R \iff \pi \circ x \in S$ for all $x \in X^d$. The *saturation* of a set $Y \subseteq X$ under a bijection $T: X \rightarrow X$ is given by $[Y]_T = \bigcup_{n \in \mathbb{Z}} T^n[Y]$. A subset of a topological space is Π_2^0 if it is a countable intersection of open sets, and Σ_3^0 if it is a countable union of Π_2^0 sets. The primary new technical result underlying our proof of Theorem 1 is the following:

Theorem 2 (ZF + DC). *Suppose that $d \geq 3$ and G is a Σ_3^0 graph on $2^{\mathbb{N}}$. Then there is a continuous embedding $\pi: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ of $\text{Hyp}^d(G)$ into the d -ary Li–Yorkeness relation of a homeomorphism $T: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ for which there is a unique point of $2^{\mathbb{N}}$ not in the T -saturation of $\pi[2^{\mathbb{N}}]$.*

In §1, we establish Theorem 2. In §2, we obtain Theorem 1 and a related independence phenomenon as corollaries.

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1. EMBEDDING HYPERGRAPHS

Define $S: 2^{\mathbb{Z}} \rightarrow 2^{\mathbb{Z}}$ by $S(c)(i) = c(i+1)$ for all $i \in \mathbb{Z}$ and $c \in 2^{\mathbb{Z}}$. Theorem 2 follows from [Kec95, Proposition 7.4] and:

Theorem 1.1 (ZF + DC). *Suppose that $d \geq 3$ and G is a Σ_3^0 graph on $2^{\mathbb{N}}$. Then there exist an S -invariant perfect set $F \subseteq 2^{\mathbb{Z}}$, whose finite sequences are proximal, and a continuous embedding $\pi: 2^{\mathbb{N}} \rightarrow F$ of $\text{Hyp}^d(G)$ into the d -ary separation relation for which $0^{\mathbb{Z}}$ is the unique point of F not in the S -saturation of $\pi[2^{\mathbb{N}}]$.*

Proof. For all $s \in 2^{<\mathbb{N}}$, define $\mathcal{N}_s = \{c \in 2^{\mathbb{N}} \mid \forall i < |s| \ s(i) = c(i)\}$. For all $S \subseteq 2^{<\mathbb{N}}$, define $\mathcal{N}_S = \bigcup_{s \in S} \mathcal{N}_s$. For all $\ell \in \mathbb{N}$ and $p: 2^\ell \rightarrow d$, define $\ell_p = \ell$ and $\mathcal{N}_p = \prod_{i < d} \mathcal{N}_{p^{-1}(\{i\})}$. We say that p is a *near bijection* if it is surjective and there is a unique $k_p < d$ for which $|p^{-1}(\{k_p\})| > 1$. For all $k \in d \setminus \{k_p\}$, let $p^{-1}(k)$ denote the unique element of $p^{-1}(\{k\})$.

Fix Π_2^0 graphs G_n on $2^{\mathbb{N}}$ for which $G = \bigcup_{n \in \mathbb{N}} G_n$, and decreasing sequences $(G_{m,n})_{m \in \mathbb{N}}$ of open graphs whose intersection is G_n . Set $H_{m,n} = \text{Hyp}^d(G_{m,n})$ and $H_n = \bigcap_{m \in \mathbb{N}} H_{m,n}$. Then $\text{Hyp}^d(G) = \bigcup_{n \in \mathbb{N}} H_n$. For all $m, n \in \mathbb{N}$, let $P_{m,n}$ denote the set of all near bijections p for which $\ell_p \geq m$, $p(0^{\ell_p}) = 0$, and $\mathcal{N}_p \subseteq H_{m,n}$. Let $Q_{m,n}$ denote the set of all $q \in P_{m,n}$ for which there is no $p \in P_{m,n}$ such that $k_p = k_q$, $\ell_p < \ell_q$, and $p^{-1}(k) = q^{-1}(k) \upharpoonright \ell_p$ for all $k \in d \setminus \{k_p\}$.

Fix an infinite set $I \subseteq \mathbb{N} \setminus \{0\}$ such that $|I \cap (I - n)| < \aleph_0$ for all $n > 0$, and a bijection $\phi: I \rightarrow \mathbb{N} \cup \bigcup_{m,n \in \mathbb{N}} \{m\} \times \{n\} \times Q_{m,n}$ such that $I \cap [i, 2i + (d-2)(\phi(i)(1) + 1)] = \{i\}$ for all $i \in I \setminus \phi^{-1}(\mathbb{N})$. Define a continuous function $\pi: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{Z}}$ by setting $\pi(c)(i) = 1$ if and only if

- (1) $i = 0$;
- (2) $i \in I$, $\phi(i) \in \mathbb{N}$, and $c(\phi(i)) = 1$; or
- (3) $\exists 0 < k < d \exists m, n \in \mathbb{N} \exists q \in Q_{m,n}$
 $((m, n, q) = \phi(i - (k-1)(n+1)) \text{ and } q(c \upharpoonright \ell_q) = k)$.

Lemma 1.2. *The function π is injective.*

Proof. Suppose that $c, c' \in 2^{\mathbb{N}}$ are distinct. Then there exists $n \in \mathbb{N}$ with $c(n) \neq c'(n)$. Fix $i \in I$ for which $\phi(i) = n$. As $\pi(c)(i) = c(n)$ and $\pi(c')(i) = c'(n)$, it follows that $\pi(c)(i) \neq \pi(c')(i)$, so $\pi(c) \neq \pi(c')$. $\square$

The *product metric* on $2^{\mathbb{Z}}$ is given by $\rho(c, c') = \sum_{i \in \mathbb{Z}} |c(i) - c'(i)|/2^{|i|}$ for all $c, c' \in 2^{\mathbb{Z}}$. A *homomorphism* from a d -ary relation R on a set Y to a d -ary relation S on a set Z is a function $\pi: Y \rightarrow Z$ such that $\pi \circ y \in S$ for all $y \in R$. Define $N_i = \{i\}$ for all $i \in \phi^{-1}(\mathbb{N})$ and $N_i = [i, i + (d-2)(\phi(i)(1) + 1)]$ for all $i \in I \setminus \phi^{-1}(\mathbb{N})$.

Lemma 1.3. *The function π is a homomorphism from $\text{Hyp}^d(G)$ to the d -ary separation relation.*

Proof. Suppose that $c \in \text{Hyp}^d(G)$, and fix $n \in \mathbb{N}$ and $j, k < d$ such that $c(j) G_n c(k)$. By permuting the coordinates of c , we can assume that either $c(0) = 0^{\mathbb{N}}$ or both $0^{\mathbb{N}} \notin c[d]$ and $j, k > 0$. Fix any $k' \in d \setminus \{j, k\}$ in the former case, and set $k' = 0$ in the latter. If $m \in \mathbb{N}$ is sufficiently large that $\text{proj}_{2^m} \circ c$ is injective and $0^{\mathbb{N}} \notin c[d] \implies 0^m \notin (\text{proj}_{2^m} \circ c)[d]$, then there is a least $\ell \geq m$ for which there is a near bijection $q \in P_{m,n}$ extending $(\text{proj}_{2^\ell} \circ c)^{-1}$ such that $k_q = k'$ and $\ell_q = \ell$. As $q \in Q_{m,n}$, there exists $i \in I$ with $\phi(i) = (m, n, q)$. Then $\text{proj}_{2^{N_i}} \circ \pi \circ c$ is injective, so $\min_{j < k < d} \rho(S^i(\pi(c(j))), S^i(\pi(c(k)))) \geq 1/2^{(d-2)(n+1)}$. $\square$

Lemma 1.4. *The function π is a homomorphism from $(2^{\mathbb{N}})^d \setminus \text{Hyp}^d(G)$ to the complement of the d -ary separation relation.*

Proof. Suppose that $c \in (2^{\mathbb{N}})^d$ and $\pi \circ c$ is separated. Fix $\epsilon > 0$ such that $I' = \{i' \in \mathbb{N} \mid \min_{j < k < d} \rho(S^{i'}(\pi(c(j))), S^{i'}(\pi(c(k)))) \geq \epsilon\}$ is infinite, as well as $r \in \mathbb{N}$ such that $\sum_{|i| > r} 1/2^{|i|} < \epsilon$. By throwing out finitely many elements of I' , we can assume that for all $i' \in I'$, there is at most one $i \in I$ such that $[i' - r, i' + r] \cap N_i \neq \emptyset$. Our choice of r then yields a unique such $i \in I$, in which case $\text{proj}_{2^{N_i}} \circ \pi \circ c$ is injective. As $d \geq 3$, there exist $m, n \in \mathbb{N}$, $q \in Q_{m,n}$, and $\sigma \in \text{Sym}(d)$ for which $(d-2)(n+1) \leq 2r$, $\phi(i) = (m, n, q)$, and $c \circ \sigma \in \mathcal{N}_q$. By passing to an infinite subset of I' , we can assume that the values of k_q , n , and σ are independent of $i' \in I'$. The minimality in the definition of $Q_{m,n}$ from $P_{m,n}$ then ensures that elements of I' corresponding to the same value of m also correspond to the same value of i . As the distance between any two such elements of I' is at most $2r$, it follows that $c \in H_{m,n}$ for infinitely many $m \in \mathbb{N}$, so $c \in H_n$, thus $c \in \text{Hyp}^d(G)$. $\square$

The closure F of the S -saturation of $\pi[2^{\mathbb{N}}]$ is clearly perfect.

Lemma 1.5. *The sequence $0^{\mathbb{Z}}$ is the unique element of F not in $[\pi[2^{\mathbb{N}}]]_S$.*

Proof. Suppose that $c \in F \setminus [\pi[2^{\mathbb{N}}]]_S$. Then there exist a sequence $(c_n)_{n \in \mathbb{N}}$ of elements of $2^{\mathbb{N}}$, $\epsilon \in \{\pm 1\}$, and a strictly increasing sequence $(i_n)_{n \in \mathbb{N}}$ of natural numbers such that $S^{\epsilon i_n}(\pi(c_n)) \rightarrow c$. If $\epsilon = -1$, then clearly $c = 0^{\mathbb{Z}}$. If $\epsilon = 1$ and $c \neq 0^{\mathbb{Z}}$, then our choice of I ensures that c is in the S -orbit of $\pi(0^{\mathbb{N}})$, a contradiction. $\square$

Lemma 1.5 implies that the union of the supports of finitely many elements of F is contained in a finite union of translates of $\{0\} \cup \bigcup_{i \in I} N_i$. As the defining property of I ensures that such sets have arbitrarily long gaps, every finite sequence of elements of F is proximal. $\square$

A set $Y \subseteq X$ is a G -clique if all distinct elements of Y are G -related, and G -independent if no elements of Y are G -related.

Proposition 1.6 (ZF + DC). *Suppose that $d \geq 2$, X and Y are Polish spaces, G is a Borel graph on X , $T: Y \rightarrow Y$ is a Borel automorphism, $B \subseteq Y$ is an uncountable Borel d -scrambled set, $\pi: X \rightarrow Y$ is an injective Borel homomorphism from $X^d \setminus \text{Hyp}^d(G)$ to the complement of the d -ary Li–Yorkeness relation, and the saturation of $\pi[X]$ is co-countable. Then G has a perfect clique.*

Proof. Fix $i \in \mathbb{Z}$ for which $B \cap (T^i \circ \pi)[X]$ is uncountable. The perfect set theorem and Galvin’s Theorem (see, for example, [Kec95, Theorems 13.6 and 19.7]) then yield a perfect set $P \subseteq (T^i \circ \pi)^{-1}(B)$ that is either a G -clique or G -independent. But G -independence would contradict the fact that B is d -scrambled. $\square$

2. D -SCRAMBLED SETS

A subset of a topological space is Σ_2^0 if its complement is Π_2^0 .

Proof (of Theorem 1). By [KV12, Theorem 2.1] and a simple Baire category argument, there is a Σ_2^0 graph G on $2^{\mathbb{N}}$ with an uncountable clique $Y \subseteq 2^{\mathbb{N}}$ but no perfect clique. By Theorem 2, there is a continuous embedding $\pi: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ of $\text{Hyp}^d(G)$ into the d -ary Li–Yorkeness relation of a homeomorphism $T: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ for which the saturation of $\pi[2^{\mathbb{N}}]$ is co-countable. Then $\pi[Y]$ is an uncountable d -scrambled set, but there is no perfect d -scrambled set by Proposition 1.6. $\square$

Let $\mathfrak{c}$ denote the cardinality of the continuum.

Theorem 2.1 (ZFC + $\neg\text{CH}$). *Suppose that $d \geq 3$. Then it is consistent that there is a homeomorphism $T: 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$ with a d -scrambled set of cardinality $\mathfrak{c}$ but no perfect d -scrambled set.*

Proof. By [She99, Theorem 1.13], we can assume that there is a Σ_2^0 graph G on $2^{\mathbb{N}}$ with a clique of cardinality $\mathfrak{c}$ but no perfect clique. The proof of Theorem 1 therefore yields the desired result. $\square$

A *hypergraph* is a permutation-invariant set of injective sequences.

Theorem 2.2 (ZFC + $\neg\text{CH}$). *Suppose that $d \geq 3$. Then it is consistent that every Borel function on a Polish space with a d -scrambled set of cardinality $\aleph_2$ has a d -scrambled perfect set.*

Proof. By [KS03, Proposition 3.4], we can assume that every Borel d -ary hypergraph on a Polish space with a clique of cardinality $\aleph_2$ has a perfect clique. But d -ary Li–Yorkeness is a Borel hypergraph. $\square$

Remark. For all $D \subseteq \mathbb{N} \setminus 3$, generalizations of our main results to individual sets that are simultaneously d -scrambled for all $d \in D$ can be obtained by interleaving the underlying arguments.

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